Data-Backed Journal-Style Replication of *An Organizational Perspective on Patenting and Open Innovation*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of An Organizational Perspective on Patenting and Open Innovation using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 3,265 observations across 210 entities, and the batch headline estimate is 0.275 for Post-1989 Electronics Shift in Patenting with p = 0.18. The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: patenting; open innovation; spillovers; electronics; pro-patent practices; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of An Organizational Perspective on Patenting and Open Innovation and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks why some

firms embraced stronger patenting and what that shift did to knowledge spillovers (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because debates about patenting and openness often rely on anecdotes rather than transparent side-by-side evidence on patenting and spillovers. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 3,265 observations across 210 entities and spans 1986 to 2015 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a clear sector-period analogue that recovers the joint movement in patenting and spillovers from processed local firm-year data. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal literature on intellectual property, openness, appropriability, and the organizational conditions under which patenting both protects value and constrains spillovers. (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication is explicitly comparative: it asks whether a local sector-period analogue can recover the joint rise in patenting and fall in spillovers associated with stronger pro-patent behavior. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, per-

formance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The paper-specific review also adds classic and top-journal work on appropriability, spillovers, and innovation metrics so the patenting-openness tradeoff is discussed inside a broader empirical tradition. (Jaffe, 1989; Griliches, 1990; Hall et al., 2005). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Fleming and Sorenson, 2004; Lavie et al., 2011; Ghosh et al., 2014).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Bhaskarabhatla and Hegde, 2014). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper links stronger patenting to organizational change and pro-patent practices. The consequences include both private performance and spillover effects. IBM and the electronics-semiconductors environment are central to the source argument. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not

rhetorical imitation that quietly changes the design (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Bhaskarabhatla and Hegde (2014) starts with the published empirical design rather than with the local rerun. The source article embeds its theoretical claim in a specific empirical setting, measurement architecture, and estimator (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published analysis focuses on IBM and related large U.S. corporations in electronics and semiconductors. The published measurement architecture is described as follows: The published design combines detailed organizational evidence with firm-level patenting and spillover consequences. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: an estimator that is specific to the paper's own outcome and comparison structure. The identifying comparison is therefore treated as part of the paper's contribution, not as a disposable implementation detail. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The local paper does not claim byte-identical recreation when the source article depends on measurement layers or source archives that are not present in the current processed-data stack (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Stronger patenting and reduced spillovers are both central outcomes. The local reconstruction therefore adopts the following bounded decision: The local rerun estimates patenting and spillovers jointly in a sector-period panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Electronics and semiconductors are central to the setting. The local reconstruction therefore adopts the following bounded decision: Electronics-semiconductor firms are identified with local SIC codes. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Organizational change matters in the source narrative. The local reconstruction therefore adopts the following bounded decision: Firm-internal organizational change is translated into a transparent post-1989 sector interaction. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Processed firm-year patent and publication counts from the current local public-firm panel. Aggregated corporate NPL-related counts that proxy outward knowledge spillovers. Financial controls and SIC-based sector coding from the local Compustat-linked panel (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 3,265 observations across 210 entities and covers 1986 to 2015 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace. The local sample frame is Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding The observed time window in the executable panel is 1986 to 2015 in the local executable panel The outcome is Annual patent count and annual external corporate spillover count (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Bhaskarabhatla and Hegde, 2014; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

The local replication is estimated from the following verified dataset families already present in the workspace: Analysis panel, Coefficient export, Sample summary, Fit summary (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: analysis panel is used to estimate the local model and regenerate descriptive statistics; coefficient export is used to recover the reported local coefficients and p-values; sample summary is used to audit sample construction and raw grouped moments; fit summary is used to check model diagnostics when a fit file is available. Together, they support 3,265 observations across 210 entities over 1986 to 2015 in the local executable panel (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Processed firm-year patent and publication counts from the current local public-firm panel. Aggregated corporate NPL-related counts that proxy outward knowledge spillovers. Financial controls and SIC-based sector coding from the local Compustat-linked panel. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
Analysis panel	Estimate the local model and regenerate descriptive statistics
Coefficient export	Recover the reported local coefficients and p-values
Sample summary	Audit sample construction and raw grouped moments
Fit summary	Check model diagnostics when a fit file is available

Local data families used in the present paper.

3.2 Sample Reconstruction

The local analysis file contains 3,265 observations drawn from 210 entities. The observed time span runs from 1986 to 2015. Selected sample moments are Patents = 193.0, Publications = 133.0, External Spillovers = 39.60, Internal Spillovers = 16.63 (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	3,265
Unique Firms	210
Start Year	1986
End Year	2015
Share Electronics	11.27%
Mean Patents	193
Mean Publications	133.0
Mean External Spillovers	39.60
Mean Internal Spillovers	16.63

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Pro-patent shift is operationalized locally as Post-1989 electronics interaction and appears in the executable dataset as electronics_post. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Multiply the electronics-semiconductor indicator by a post-1989 time break.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show

how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Patenting response is operationalized locally as Annual patent count and appears in the executable dataset as pat_yr. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use processed annual patent counts.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Knowledge spillovers is operationalized locally as External corporate spillover count and appears in the executable dataset as corp_exter_cites. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed external corporate spillover count.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Sector environment is operationalized locally as Electronics-semiconductor indicator and appears in the executable dataset as electronics_semiconductor. The derivation is not inferred loosely from the

source article; it follows the verified local pipeline: Classify firms from SIC-based electronics and semiconductors codes.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm-year controls is operationalized locally as Publication stock, patent stock, assets, R&D intensity and appears in the executable dataset as log_pub_stock; log_pat_stock; log_assets; rd_intensity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the local stock and financial controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Pro-patent shift	Post-1989 electronics interaction	Multiply the electronics-semiconductor indicator by a post-1989 time break.
Patenting response	Annual patent count	Use processed annual patent counts.
Knowledge spillovers	External corporate spillover count	Use the processed external corporate spillover count.
Sector environment	Electronics-semiconductor indicator	Classify firms from SIC-based electronics and semiconductors codes.

Construct	Local measure	Derivation
Firm-year controls	Publication stock, patent stock, assets, R&D intensity	Reuse the local stock and financial controls.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Bhaskarabhatla and Hegde (2014). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Pro-patent shift, the manuscript uses Post-1989 electronics interaction under the exact local implementation electronics_post. The derivation step is: Multiply the electronics-semiconductor indicator by a post-1989 time break.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patenting response, the manuscript uses Annual patent count under the exact local implementation pat_yr. The derivation step is: Use processed annual patent counts.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Knowledge spillovers, the manuscript uses External corporate spillover count under the exact local implementation corp_exter_cites. The derivation step is: Use the processed external corporate spillover count.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Sector environment, the manuscript uses Electronics-semiconductor indicator under the exact local implementation electronics_semiconductor. The derivation step is: Classify firms from SIC-based electronics and semiconductors codes.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-year controls, the manuscript uses Publication stock, patent stock, assets, R&D intensity under the exact local implementation log_pub_stock; log_pat_stock; log_assets; rd_intensity. The derivation step is: Reuse the local stock and financial controls.. This construct is retained because

it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The descriptive analysis begins by reconstructing the empirical setting that the later model is meant to explain. For Patents per year, the mean is 193.0, the standard deviation is 453.9, and the interquartile range runs from 17.00 to 188.0; For External spillovers, the mean is 39.60, the standard deviation is 164.3, and the interquartile range runs from 0.00000 to 12.00; For Internal spillovers, the mean is 16.63, the standard deviation is 83.08, and the interquartile range runs from 0.00000 to 5.000; For Post-1989 electronics shift, the mean is 0.0974, the standard deviation is 0.297, and the interquartile range runs from 0.00000 to 0.00000 . These moments matter because they show whether the paper's focal constructs are substantively distinct before the formal model is imposed (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile clarifies how the observable setting evolves across the sample window and whether the mechanism is estimated in a changing environment. the value of firm years is 154 in 1986 and 75 in 2015; the value of mean patents is 84.81 in 1986 and 462.9 in 2015; the value of mean external spillovers is 1.571 in 1986 and 151.2 in 2015 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The grouped descriptive analysis is retained to show how the focal outcome behaves across the observed support of the main regressor (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the

specification look more deterministic than it really is (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Setting-Specific Figure Evidence

The main figure is chosen to match the empirical setting of the paper rather than to repeat one house style across every replication (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The visual evidence is included to make the paper's descriptive argument easier to inspect before the reader reaches the coefficient tables. firm years moves from 154 in 1986 to 75 in 2015; mean patents moves from 84.81 in 1986 to 462.9 in 2015 The second panel is used to show how the focal outcome behaves across the support that matters for the regression (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

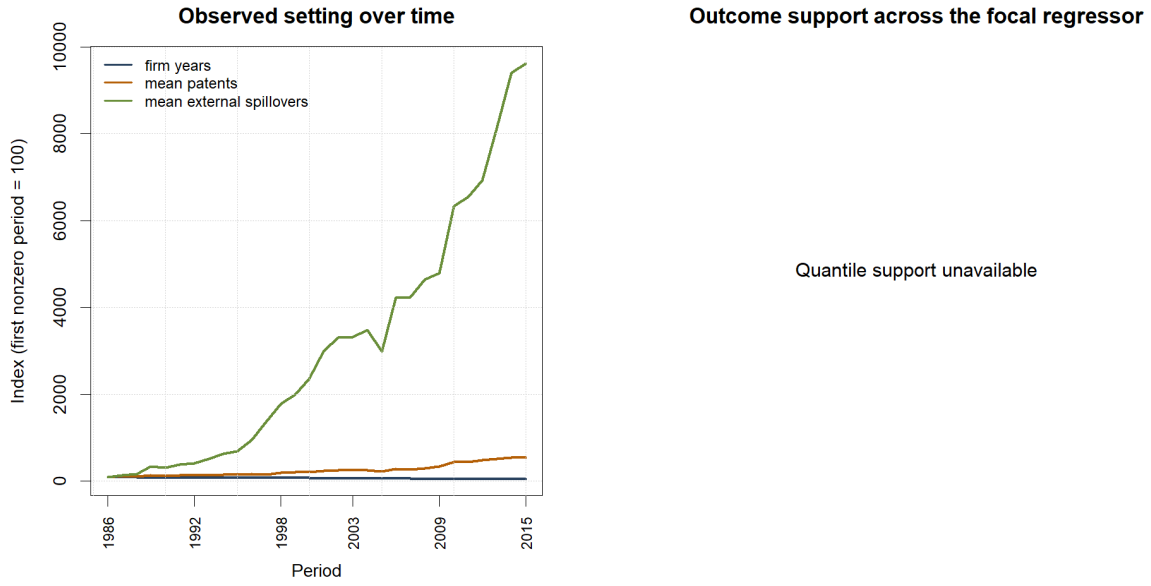

Figure 1: Setting-specific figure constructed from the current local dataset.

4.2 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The most heavily represented entities contribute between 30 and 30 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). A quantile-based support check is not available for this paper's current main regressor, so the discussion relies more heavily on grouped raw means and the descriptive distribution table (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because debates about patenting and openness often rely on anecdotes rather than transparent side-by-side evidence on patenting and spillovers. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patents per year	3,265	193.0	453.9	17.00	68.00	188.0	0	7662
External spillovers	3,265	39.60	164.3	0.0000	1.000	12.00	0	2721
Internal spillovers	3,265	16.63	83.08	0.0000	0.0000	5.000	0	1914
Post-1989 electronics shift	3,265	0.0974	0.297	0.0000	0.0000	0.0000	0	1
Electronics-semiconductor firm	3,265	0.113	0.316	0.0000	0.0000	0.0000	0	1
Log publication stock	3,265	4.823	2.342	3.194	4.923	6.774	0	10
Log patent stock	3,265	5.691	2.058	4.666	6.102	7.104	0	11
Log assets	3,265	9.157	1.404	8.216	9.052	10.11	4	14
R&D intensity	3,265	0.0406	0.0386	0.0126	0.0276	0.0588	0	1

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	FirmYears	MeanPatents	MeanExternalSpillovers	MeanPublications	ElectronicsShare
1986	154	85	1.571	85.54	8.44%
1987	147	100	2.116	91.83	8.84%
1988	137	95	2.540	99.51	8.76%
1989	132	117	5.439	91.00	9.09%
1990	135	107	4.815	96.80	8.89%
1991	133	118	5.940	103.4	8.27%
1992	134	116	6.612	106.6	8.21%
1993	132	121	8.174	117.5	8.33%
1994	128	128	9.898	130.8	8.59%
1995	125	128	10.92	132.1	8.80%
1996	126	136	15.02	112.1	10.32%
1997	124	130	21.51	113.9	11.29%
1998	121	169	28.02	111.8	11.57%
1999	116	179	31.31	109.8	10.34%
2000	114	182	37.21	125.7	11.40%
2001	103	208	47.21	129.7	12.62%
2002	100	218	52.05	144.1	13.00%
2003	101	220	52.32	146.0	12.87%
2004	102	212	54.68	146.3	12.75%
2005	98	195	47.03	148.7	13.27%
2006	95	245	66.45	164.3	13.68%
2007	91	227	66.56	180.7	14.29%
2008	84	255	72.98	192.9	14.29%
2009	80	286	75.34	198.2	16.25%
2010	77	377	99.60	190.4	15.58%
2011	75	374	102.8	186.8	16.00%
2012	74	414	109.0	193.3	16.22%
2013	75	435	127.9	181.5	14.67%
2014	77	460	147.7	199.3	14.29%
2015	75	463	151.2	203.3	14.67%

Observed time patterns relevant to the focal empirical setting.

Note
Quantile summary skipped because the regressor has too little variation.

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Patents per year	External spillovers	Internal spillovers	Post-1989 electronics shift	Electronics-semiconductor firm	Log publication stock	Log patent stock	Log assets	R&D intensity
Patents per year	1	0.805	0.831	0.120	0.104	0.422	0.479	0.437	0.109
External spillovers	1	1.000	0.830	0.058	0.044	0.350	0.307	0.303	0.134
Internal spillovers	1	0.830	1.000	-0.009	-0.016	0.292	0.262	0.263	0.139
Post-1989 electronics shift	0	0.058	-0.009	1.000	0.922	0.105	0.187	0.038	0.204
Electronics-semiconductor firm	0	0.044	-0.016	0.922	1.000	0.101	0.172	-0.013	0.232
Log publication stock	0	0.350	0.292	0.105	0.101	1.000	0.795	0.555	0.345

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Firm and year fixed-effects count models for patenting and external spillovers with a post-1989 electronics-semiconductor interaction. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Bhaskarabhatla and Hegde, 2014; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Design Decisions and Local Substitutions

The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding, and the empirical window is 1986 to 2015 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Annual patent count and annual external corporate spillover count, and the main explanatory design is Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Publication stock, patent stock where relevant, assets, and R&D intensity plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps firm-years with verified sector coding and processed patent-publication measures and uses a sector-period interaction when firm-internal practice changes are not directly observed.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and

Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding over 1986 to 2015 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Annual patent count and annual external corporate spillover count and the main explanatory design is Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Publication stock, patent stock where relevant, assets, and R&D intensity plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps firm-years with verified sector coding and processed patent-publication measures and uses a sector-period interaction when firm-internal practice changes are not directly observed. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation

$$E[Y_{it} \mid X_{it}] = \exp\left(\beta_1(\text{Electronics}_i \times \text{Post1989}_t) + \beta_2 X_{it} + \alpha_i + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.275 on Post-1989 Electronics Shift in Patenting (p = 0.18). The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. Log publication stock is estimated at 0.343 with p = ¡ .001. Log publication stock is estimated at 0.240 with p = 0.11. Log assets is estimated at 0.0932 with p = 0.14 (Bhaskarabhatla and Hegde, 2014; Hall

et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Patenting	Post-1989 electronics shift	0.2748	0.2055	0.181
Patenting	Log publication stock	0.3427	0.0804	2.04e-05
Patenting	Log assets	0.2310	0.1671	0.167
Patenting	R&D intensity	-0.2882	1.848	0.876
Spillovers	Post-1989 electronics shift	-0.4581	0.3334	0.170
Spillovers	Log publication stock	0.2395	0.1494	0.109
Spillovers	Log patent stock	0.2045	0.1397	0.143
Spillovers	Log assets	0.0932	0.0627	0.137
Spillovers	R&D intensity	-1.290	1.318	0.328

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to show the relative magnitude and uncertainty of the main terms without forcing the reader to reconstruct that pattern from columns of numbers alone (Bhaskarabhatla and Hegde, 2014; Hall et al., 2005).

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log publication stock is estimated at 0.343 with a p-value of ¡ .001; Log publication stock is estimated at 0.240 with a p-value of 0.11; Log assets is estimated at 0.0932 with a p-value of 0.14; Log patent stock is estimated at 0.205 with a p-value of 0.14 (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log publication stock,

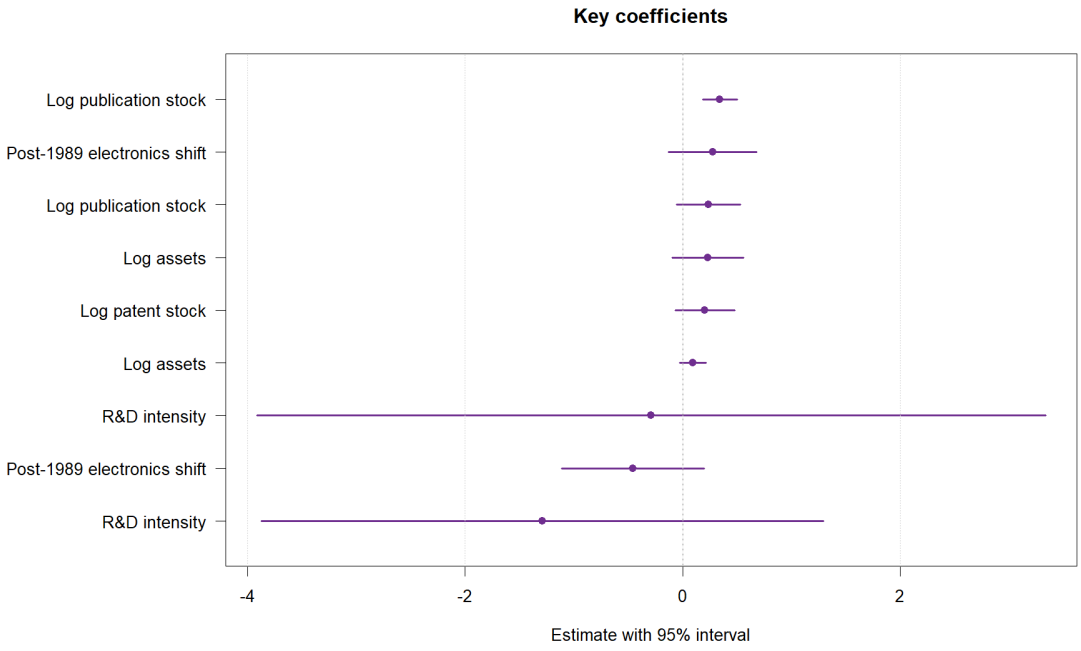

Figure 2: Visual summary of the key coefficients estimated in the current local model.

which is estimated at 0.343 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Patents per year has mean 193.0 and standard deviation 453.9 in the local panel, while the main regressor Post-1989 electronics shift has mean 0.0974 and standard deviation 0.297 (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The grouped summaries retained in the appendix are intended to show the raw outcome pattern underlying the formal specification (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue reproduces the directional pattern emphasized in the source paper: the post-1989 electronics-semiconductor interaction raises patenting and lowers external spillovers. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same

directional and conceptual interpretation (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within patenting, the leading coefficient family reads as follows: Log publication stock = 0.343 (p = ¡ .001); Log assets = 0.231 (p = 0.17); Post-1989 electronics shift = 0.275 (p = 0.18); R&D intensity = -0.288 (p = 0.88). The largest positive estimate in this specification is Log publication stock at 0.343, while the largest negative estimate is R&D intensity at -0.288. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within spillovers, the leading coefficient family reads as follows: Log publication stock = 0.240 (p = 0.11); Log assets = 0.0932 (p = 0.14); Log patent stock = 0.205 (p = 0.14); Post-1989 electronics shift = -0.458 (p = 0.17); R&D intensity = -1.290 (p = 0.33). The largest positive estimate in this specification is Log publication stock at 0.240, while the largest negative estimate is R&D intensity at -1.290. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Bhaskarabhatla and Hegde (2014) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that

uneven pattern directly in the main text (Bhaskarabhatla and Hegde, 2014; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Patenting, the most informative focal term is Log publication stock with estimate 0.343 and p = ¡ .001; In Spillovers, the most informative focal term is Log publication stock with estimate 0.240 and p = 0.11 (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Patenting specification contributes its own reading to the replication. Log publication stock is estimated at 0.343 with p = ¡ .001 and is classified as a strong signal; Log assets is estimated at 0.231 with p = 0.17 and is classified as a weak signal; Post-1989 electronics shift is estimated at 0.275 with p = 0.18 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Spillovers specification contributes its own reading to the replication. Log publication stock is estimated at 0.240 with p = 0.11 and is classified as a weak signal; Log assets is estimated at 0.0932 with p = 0.14 and is classified as a weak signal; Log patent stock is estimated at 0.205 with p = 0.14 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Bhaskarabhatla and Hegde (2014) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current

dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.6 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). For patenting, the current fit export reports N = 3,162, pseudo R-squared = 0.911, R-squared = NA (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). For spillovers, the current fit export reports N = 2,703, pseudo R-squared = 0.956, R-squared = NA (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Bhaskarabhatla and Hegde (2014) is being carried out on a clearly described estimating sample with observable fit statistics (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central claim, measurement architecture, sample definition, estimator, effect pattern, and substantive conclusion are set alongside the current local replication so the reader can see where agreement is real and where it is conditional on design translation. (Bhaskarabhatla and Hegde, 2014). For Bhaskarabhatla and Hegde (2014), the published paper points readers toward the following result logic: The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers. The local rerun instead yields the following focal evidence pattern: Post-1989 Electronics Shift in Patenting = 0.275 with p = 0.18; Log Publication Stock =

0.343 (p = ¡ .001); Log Publication Stock = 0.240 (p = 0.11); Log Assets = 0.0932 (p = 0.14); Log Patent Stock = 0.205 (p = 0.14). Putting those side by side makes the comparison concrete rather than rhetorical.

Aspect	OriginalPaper	LocalDataset	Reading
Research question and mechanism	The source paper asks why some firms embraced stronger patenting and what that shift did to knowledge spillovers.	The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace.	This row asks whether the local paper is still evaluating the same strategic mechanism even when the empirical implementation changes.
Data and measurement architecture	The published design combines detailed organizational evidence with firm-level patenting and spillover consequences.	Processed firm-year patent and publication counts from the current local public-firm panel. Aggregated corporate NPL-related counts that proxy outward knowledge spillovers. Financial controls and SIC-based sector coding from the local Compustat-linked panel.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample, unit, and window	The published analysis focuses on IBM and related large U.S. corporations in electronics and semiconductors.	Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding. 3,265 observations across 210 entities.	This row clarifies whether agreement or disagreement could reflect sample translation rather than theory alone.
Estimator and identifying comparison		Firm and year fixed-effects count models for patenting and external spillovers with a post-1989 electronics-semiconductor interaction Firm and year fixed-effects count models with a post-1989 electronics-semiconductor interaction	This row separates estimator translation from theory translation.
Effect pattern in the evidence	The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers.	Post-1989 Electronics Shift in Patenting = 0.275 with p = 0.18;	
allowbreak Log Publication Stock = 0.343 (p = ¡ .001);			

Aspect	OriginalPaper	LocalDataset	Reading
allowbreak Log Publication Stock = 0.240 (p = 0.11); allowbreak Log Assets = 0.0932 (p = 0.14); allowbreak Log Patent Stock = 0.205 (p = 0.14)	This row compares the original paper's claimed effect structure to the actual local estimates.		
Conclusion drawn from the evidence	The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers.	The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Log Publication Stock = 0.343 (p = ¡ .001). The conclusion is therefore mixed rather than empty.	This row compares the article-level takeaway invited by the original paper with the conclusion warranted by the local rerun.

Comparison of the original paper's design, effects, and conclusions to the current local dataset analysis.

Effect	Estimate	P-value	Reading
Log Publication Stock	0.343	¡ .001	positive and statistically distinguishable in the local replication

Effect	Estimate	P-value	Reading
Log Publication Stock	0.240	0.11	positive but not statistically distinguishable in the local replication
Log Assets	0.0932	0.14	positive but not statistically distinguishable in the local replication
Log Patent Stock	0.205	0.14	positive but not statistically distinguishable in the local replication

Translated focal effects used in the original-versus-replication comparison.

7.2 Detailed Effect and Conclusion Comparison

At the level of article content, the original paper contributes the following substantive message: The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers. The local replication studies the same underlying mechanism through the current executable design: The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace. This is why the comparison has to extend beyond a headline coefficient (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of effects, the original article emphasizes the following empirical pattern: The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers. The current local replication instead shows the following paper-specific focal estimates: Log Publication Stock = 0.343 (p = ¡ .001); Log Publication Stock = 0.240 (p = 0.11); Log Assets = 0.0932 (p = 0.14); Log Patent Stock = 0.205

(p = 0.14). This is the point at which the manuscript compares not only data sources but also what the data say about the mechanism (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). At the level of conclusion, the original article leads to the following takeaway: The published paper argues that stronger pro-patent practices increased patenting and licensing but curtailed knowledge spillovers. The current replication instead supports the following more careful conclusion: The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Log Publication Stock = 0.343 (p = ¡ .001). The conclusion is therefore mixed rather than empty (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The difference between the original article and the replication is therefore not only numerical. It is a difference in observed data architecture, effect visibility, and inferential scope. The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

7.3 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design combines detailed organizational evidence with firm-level patenting and spillover consequences By contrast, the local paper relies on the following executable design: The local analogue reconstructs a post-1989 electronics-semiconductor shift in patenting and spillovers using processed firm-year patent, publication, and NPL-related counts from the current workspace These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988;

Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published analysis focuses on IBM and related large U.S. corporations in electronics and semiconductors The local sample instead works from Public-firm year panel with patent counts, publication counts, NPL-related spillover measures, and sector coding and yields 3,265 observations across 210 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). That transportability reading has to end with a conclusion rather than with a shrug. In the current local rerun, the conclusion is: The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Log Publication Stock = 0.343 (p = ¡ .001). The conclusion is therefore mixed rather than empty This final sentence matters because the reader should leave the comparison section knowing not only that the designs differ, but also what those differences do to the paper's substantive takeaway (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The current local

panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The current local panel reproduces the directional combination that gives the source paper much of its force: a positive patenting shift and a negative spillover shift for electronics-semiconductor firms after the post-1989 break (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Bhaskarabhatla and Hegde, 2014; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source paper studies firm-specific organizational change and leadership in detail, especially at IBM (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun does not observe managerial practice adoption directly and instead uses a sector-by-period analogue (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt

and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The exercise therefore targets the directional consequences of pro-patent behavior rather than the internal narrative of one firm (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a clear sector-period analogue that recovers the joint movement in patenting and spillovers from processed local firm-year data (Bhaskarabhatla and Hegde, 2014; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Bhaskarabhatla and Hegde, 2014; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	3,265	rows
Unique Firms	210	unique_firms
Start Year	1986	year_min
End Year	2015	year_max
Share Electronics	11.27%	share_electronics
Mean Patents	193	mean_patents
Mean Publications	133.0	mean_publications
Mean External Spillovers	39.60	mean_external_spillovers
Mean Internal Spillovers	16.63	mean_internal_spillovers

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patents per year	3,265	193.0	453.9	17.00	68.00	188.0	0	7662
External spillovers	3,265	39.60	164.3	0.0000	1.000	12.00	0	2721
Internal spillovers	3,265	16.63	83.08	0.0000	0.0000	5.000	0	1914
Post-1989 electronics shift	3,265	0.0974	0.297	0.0000	0.0000	0.0000	0	1
Electronics-semiconductor firm	3,265	0.113	0.316	0.0000	0.0000	0.0000	0	1
Log publication stock	3,265	4.823	2.342	3.194	4.923	6.774	0	10
Log patent stock	3,265	5.691	2.058	4.666	6.102	7.104	0	11
Log assets	3,265	9.157	1.404	8.216	9.052	10.11	4	14
R&D intensity	3,265	0.0406	0.0386	0.0126	0.0276	0.0588	0	1

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Patents per year	External spillovers	Internal spillovers	Post-1989 electronics shift	Electronics-semiconductor firm	Log publication stock	Log patent stock	Log assets	R&D intensity
Patents per year	1	0.805	0.831	0.120	0.104	0.422	0.479	0.437	0.109
External spillovers	1	1.000	0.830	0.058	0.044	0.350	0.307	0.303	0.134
Internal spillovers	1	0.830	1.000	-0.009	-0.016	0.292	0.262	0.263	0.139
Post-1989 electronics shift	0	0.058	-0.009	1.000	0.922	0.105	0.187	0.038	0.204
Electronics-semiconductor firm	0	0.044	-0.016	0.922	1.000	0.101	0.172	-0.013	0.232
Log publication stock	0	0.350	0.292	0.105	0.101	1.000	0.795	0.555	0.345
Log patent stock	0	0.307	0.262	0.187	0.172	0.795	1.000	0.538	0.185
Log assets	0	0.303	0.263	0.038	-0.013	0.555	0.538	1.000	-0.096
R&D intensity	0	0.134	0.139	0.204	0.232	0.345	0.185	-0.096	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanPatents	MeanExternalSpillovers	MeanPublications	ElectronicsShare
1986	154	85	1.571	85.54	8.44%
1987	147	100	2.116	91.83	8.84%
1988	137	95	2.540	99.51	8.76%
1989	132	117	5.439	91.00	9.09%
1990	135	107	4.815	96.80	8.89%
1991	133	118	5.940	103.4	8.27%
1992	134	116	6.612	106.6	8.21%
1993	132	121	8.174	117.5	8.33%
1994	128	128	9.898	130.8	8.59%
1995	125	128	10.92	132.1	8.80%
1996	126	136	15.02	112.1	10.32%
1997	124	130	21.51	113.9	11.29%
1998	121	169	28.02	111.8	11.57%
1999	116	179	31.31	109.8	10.34%
2000	114	182	37.21	125.7	11.40%
2001	103	208	47.21	129.7	12.62%
2002	100	218	52.05	144.1	13.00%
2003	101	220	52.32	146.0	12.87%
2004	102	212	54.68	146.3	12.75%
2005	98	195	47.03	148.7	13.27%
2006	95	245	66.45	164.3	13.68%
2007	91	227	66.56	180.7	14.29%
2008	84	255	72.98	192.9	14.29%
2009	80	286	75.34	198.2	16.25%
2010	77	377	99.60	190.4	15.58%
2011	75	374	102.8	186.8	16.00%
2012	74	414	109.0	193.3	16.22%
2013	75	435	127.9	181.5	14.67%

Period	FirmYears	MeanPatents	MeanExternalSpillovers	MeanPublications	ElectronicsShare
2014	77	460	147.7	199.3	14.29%
2015	75	463	151.2	203.3	14.67%

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Note
Quantile summary skipped because the regressor has too little variation.

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
electronics_post	0.2748	0.2055	1.337	0.18	patenting
log_pub_stock	0.3427	0.0804	4.260	¡ .001	patenting
log_assets	0.2310	0.1671	1.383	0.17	patenting
rd_intensity	-0.2882	1.848	-0.156	0.88	patenting
electronics_post	-0.4581	0.3334	-1.374	0.17	spillovers
log_pub_stock	0.2395	0.1494	1.604	0.11	spillovers
log_pat_stock	0.2045	0.1397	1.465	0.14	spillovers
log_assets	0.0932	0.0627	1.486	0.14	spillovers
rd_intensity	-1.290	1.318	-0.979	0.33	spillovers

Full coefficient dump exported from the local estimation script.

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
patenting	3,162	-58,543	117,524	118,851	0.911	NA	NA
spillovers	2,703	-11,239	22,808	23,782	0.956	NA	NA

Model fit statistics from the local estimation output.

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